always carried on in the friendliest tone. Ostwald, by his very nature a firm believer in systematization, distinguished three different types of energy, corresponding to the three spatial dimensions, namely: Distance Energy, Surface Energy, and Space Energy. Distance Energy, according to him, was the force of gravitation; Surface Energy, the surface tension of liquids; and Space Energy, the volume energy. I replied, among other comments, that there was no such thing as a volume energy in the sense specified by Ostwald. For instance, the energy of an ideal gas does not in fact even depend on the volume, but on the temperature of the gas. If an ideal gas is made to expand without doing any work, its volume increases, but the energy remains unchanged, whereas according to Ostwald, its energy ought to decrease with the decrease of the pressure.

Another controversy arose with relation to the question of the analogy between the passage of heat from a higher to a lower temperature and the sinking of a weight from a greater to a smaller height. I had emphasized the need for a sharp distinction between these two processes, for they differed from each other as basically as did the First and Second Laws of Thermodynamics. However, this theory of mine was contradicted by the view universally accepted in those days, and I just could not make my fellow physicists see it my way. In fact, certain physicists actually regarded Clausius' reasoning as unnecessarily complicated and even confused; and they refused, in particular, to admit the concept of irreversibility, and thereby to assign to heat a special position among the forms of energy. They created in opposition to Clausius' theory of thermodynamics, the so-called science of "Energetics." The first fundamental proposition of Energetics, exactly like that of Clausius' theory, expresses the principle of the conservation of energy; but its second proposition, which is supposed to formulate the direction of all occurrences, postulates a perfect analogy between the passing of heat from a higher to a lower temperature and the sinking of a weight from a greater to a smaller height. A consequence of this point of view was that the assumption of irreversibility for proving the Second Law of Thermodynamics was declared to be unessential; furthermore, the existence of an absolute zero of temperature was disputed, on the ground that for temperature, just as for height, only differences can be measured.

It is one of the most painful experiences of my entire scientific life that I have but seldom—in fact, I might say, never—succeeded in gaining universal recognition for a new result, the truth of which I could demonstrate by a conclusive, albeit only theoretical proof. This is what happened this time, too. All my sound arguments fell on deaf ears. It was simply impossible to be heard against the authority of men like Ostwald, Helm and Mach. I was firmly convinced that my claim of the basic difference between the transmission of heat and the sinking of a weight would eventually be proved to be right. But